Guangyuan Weng

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RESEARCH INTERESTS

Multimodal AI; LLM evaluation and mechanistic interpretability; recommender systems; representation learning; spatiotemporal data mining; responsible AI.

EDUCATION

Northeastern University Sept. 2021 – May 2027 expected
Ph.D. Candidate, Computer Science; Advisor: Prof. Esteban Moro *Boston, MA*

ShanghaiTech University Sept. 2017 – July 2021
B.E., Computer Science and Technology *Shanghai, China*

INDUSTRY EXPERIENCE

Chewy June – Aug. 2026
Research Scientist II Intern, MSO Data Science *Boston, MA*

- Engineered an end-to-end **multimodal GenAI pipeline** extracting structured visual and auditory semantics from video assets, integrating scene-aware keyframe sampling (FFmpeg), speech transcription (Whisper), and OCR into structured digests grounded to source timestamps.
- Developed an **LLM-as-a-judge evaluation framework** to benchmark candidate video-digest models; designed a human-in-the-loop system to discover, deduplicate, and calibrate high-dimensional creative attributes into an executable detection taxonomy.
- Formulated downstream **predictive models** (CTR and video completion) linking multimodal creative features to campaign KPIs, achieving a **15.9% lower prediction error (MAE)** over baselines across audience segments; deployed an interactive analytics dashboard presented to Chewy's **Chief Brand Officer and marketing leadership**.

Wormpex AI Research May – Aug. 2023
Research Intern; Advisor: Dr. Gang Hua (IEEE Fellow) *Bellevue, WA*

- Developed hybrid retrieval-augmented classification architectures for **long-tail visual recognition**, designing a **self-attention importance weighting module** in RAG to dynamically prioritize salient retrieved contexts and filter noise.
- Implemented and benchmarked models in **PyTorch**, building automated data ingestion, distributed training, and evaluation pipelines in close collaboration with research and engineering teams.

RESEARCH EXPERIENCE

Northeastern University, Social Urban Networks Lab Jan. 2024 – Present
Research Assistant; Advisor: Prof. Esteban Moro *Boston, MA*

LLM recommendation evaluation *Jan. 2026 – Present*

- Architected an end-to-end evaluation framework using counterfactual prompt perturbation, **dense vector embeddings**, and geospatial verification algorithms across **3 LLM families, 304 neighborhoods, and 5 U.S. metros**.
- Audited generative recommendation systems: identified a **36.8%** hallucination rate in open generation, and demonstrated that constrained retrieval eliminated fabrication while uncovering a **47.5%** systemic candidate coverage gap.

Recommender feedback simulation *June 2025 – Present*

- Developed a scalable **agent-based simulation of human–AI feedback loops**, evaluating how iterative **collaborative filtering with online retraining** shapes long-term user behavior, exposure fairness, and socioeconomic segregation.

Mobility representation learning *Jan. 2024 – June 2025*

- Developed a **mobility representation learning framework** trained on **25.4M trajectories across 11 U.S. metros**, mapping human mobility dynamics into latent metric spaces capturing behavioral connections beyond physical geography.
- Demonstrated that learned neural embeddings significantly outperform Euclidean distance in explaining urban mobility flows via gravity modeling (**Boston R^2: 0.61 vs. 0.33**), uncovering systemic socioeconomic and physical boundaries.

Northeastern University Jan. 2026 – Present
Project Leader, Mechanistic Interpretability; Advisor: Prof. David Bau *Boston, MA*

- Investigated internal representations in frontier open LLMs (**Llama-3.1-70B, Qwen-2.5-72B**) using **nsight/NDIF** for residual-stream activation caching, linear probing, and **Representational Similarity Analysis (RSA)**.
- Discovered cultural asymmetries across **2,400 global cities**; developed **contrastive activation addition (activation steering)** to intervene on latent representations and align spatial reasoning behaviors.

Northeastern University, Visual Intelligence Lab Sept. 2021 – May 2023
Research Assistant; Advisor: Prof. Huaizu Jiang *Boston, MA*

- Investigated **vision-language models (VLMs)**, few-shot learning, and compositional reasoning; designed contrastive representation learning objectives on synthetic image-graph pairs to improve transfer to real-world benchmarks.

Indiana University, Computer Vision Lab July 2020 – June 2021
Visiting Research Intern; Advisor: Prof. David J. Crandall *Bloomington, IN*

- Trained **Graph Convolutional Networks (GCNs)** to model action–object co-occurrence statistics in egocentric video; analyzed how visual priors impact out-of-distribution generalization in PyTorch.

SELECTED PUBLICATIONS AND MANUSCRIPTS

Under review

Large language models create an uneven informational layer over cities.
L. Chen, **G. Weng**, E. Moro. 2026. [Code]

Mapping the Blind Spots: Cultural Asymmetries in LLM Geographic Representations.
G. Weng, A. Malik, G. Savcisens, T. Urmi, D. Bau. 2026.

Beyond Distance: Mobility Neural Embeddings Reveal Visible and Invisible Barriers in Urban Space.
G. Weng, M. Kim, Y.-Y. Ahn, E. Moro. 2025. [Code]

Peer-reviewed publications

Action Recognition based on Cross-Situational Action-object Statistics.
S. Tsutsui, X. Wang, **G. Weng**, Y. Zhang, D. Crandall, C. Yu. *IEEE ICDL*, 2022.

Advanced mapping robot and high-resolution dataset.
H. Chen, Z. Yang, X. Zhao, **G. Weng**, et al. *Robotics and Autonomous Systems*, 2020.

Towards Generation and Evaluation of Comprehensive Mapping Robot Datasets.
H. Chen et al., including **G. Weng**. *ICRA Workshop on Dataset Generation and Benchmarking*, 2019.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, Java, Rust, SQL, R, Shell/Bash.
Machine Learning & GenAI: PyTorch, Hugging Face, JAX, vLLM, LangChain, nnsight, scikit-learn, Weights & Biases.
Core CS & Systems: Algorithms & Data Structures, Distributed Systems, Scalable System Design, Automated Pipelines.
Cloud, Tools & Platforms: Linux/Unix, Git, Docker, Amazon Web Services (AWS), Google Cloud Platform (GCP), FFmpeg, ROS.

ADDITIONAL INFORMATION

Honors & Awards: Northeastern Outstanding PhD Award for Teaching (2026); PhD Network Travel Award (2025); IU International Research Fellowship (2020); ShanghaiTech Merit Student (Top 5%, 2020).

Selected Presentations: IC2S2 2025 Plenary Talk (top 1.5% of submissions); IC2S2 2024 Plenary Talk (top 2.8%).

Academic Service: Program Committee / Reviewer for NeurIPS (2026), IC2S2 (2024–2026), and NetSci (2025–2026).

Teaching Experience: Head TA, Foundations of AI (CS5100, Fall 2025) & Mobile App Dev (CS5520, 4 terms); TA, Pattern Recognition & Computer Vision (CS5330, Fall 2021/2022).